EXPERIMENT 14

Subject-Verb Agreement Checking using CFG Feature Rules

AIM

To write a Python program that checks for number (singular/plural) agreement between subjects and verbs using Feature-Based Context-Free Grammars.

PROCEDURE

1. Define a Feature Context-Free Grammar incorporating agreement constraints (e.g. `NUM='sg'` vs `NUM='pl'`).
2. Specify lexical items associated with singular and plural feature values.
3. Initialize the FeatureChartParser with the unified feature grammar.
4. Evaluate syntactically concordant and discordant sentences.
5. Classify inputs as grammatically valid (agreement accepted) or invalid (agreement error).

PROGRAM

```
import nltk
from nltk import FeatureChartParser

fcfg_grammar = nltk.grammar.FeatureGrammar.fromstring("""
    S -> NP[NUM=?n] VP[NUM=?n]
    NP[NUM=?n] -> Det[NUM=?n] N[NUM=?n]
    VP[NUM=?n] -> V[NUM=?n] NP[NUM=?m] | V[NUM=?n]

    Det[NUM='sg'] -> 'this' | 'the' | 'a'
    Det[NUM='pl'] -> 'these' | 'the'

    N[NUM='sg'] -> 'dog' | 'cat' | 'student'
    N[NUM='pl'] -> 'dogs' | 'cats' | 'students'

    V[NUM='sg'] -> 'barks' | 'runs' | 'sleeps'
    V[NUM='pl'] -> 'bark' | 'run' | 'sleep'
""")

parser = FeatureChartParser(fcfg_grammar)

test_sentences = [
    "this dog barks",
    "these dogs bark",
    "this dog bark",
    "these dogs barks"
]

print(f"{'Sentence':<22} | {'Agreement Status':<16} | {'Parse Result'}")
print("-" * 65)
```

```
for s in test_sentences:
    tokens = s.split()
    parses = list(parser.parse(tokens))
    if len(parses) > 0:
        print(f"{s:<22} | {'VALID (Matches)':<16} | Parsed with agreement
constraint")
    else:
        print(f"{s:<22} | {'INVALID (Violation)':<16} | Agreement constraint
failed")
```

OUTPUT

```
> python3 experiment_14.py
------------------------------------------------------------------------------------------
Sentence               | Agreement Status | Parse Result
-----------------------------------------------------------------
this dog barks         | VALID (Matches)  | Parsed with agreement constraint
these dogs bark        | VALID (Matches)  | Parsed with agreement constraint
this dog bark          | INVALID (Violation) | Agreement constraint failed
these dogs barks       | INVALID (Violation) | Agreement constraint failed
[Process completed with exit code 0]
```

RESULT

Thus, subject-verb number agreement checking was successfully implemented using Feature Context-Free Grammars.